

Project 2: Cryptography MATH 4175
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For $m = 6$:
Substring $y_1 =$
LDXEJVVWTL SNY ZOLR QGT VSW QGP JXP WVR HKCKL HWPRASULXYFMKGWGTEBDQGRAFBKSLVDPZCPPQLJMJUQR
ZVEAUBVPEWAZVICGJASQRWEENLORTYCEGLILEQOMLIOEDAWLIOTOPUEWLIGKSYPOJWDWLRDEEDKWHZYM
EGVAYZZAJJEPIIYSVSFVB JLYXFDJPSHAYXWJJEZSXPLHCSNCJ
Index of Coincidence for y_1 : 0.040920483025746184

Substring $y_2 =$
DJRMMUTALSXP K MUKZIPK MDS DREI ANVHERJTNJVM LLMAKLECIVBJRGDYVLWLSLFZQRUWLRMWSPXPJKAUL
RQZSALBGTTAGQXAFVPERITFIKWLGPWRKGAEEMJGSIZZLYLVXSPPWVVRONBJWHXNEQTNFXRYMJTHRPV
AJLLKYPUEJQVLGPZWIPNZKDWKWFREHWHIZELIDDYPITUULSDIW
Index of Coincidence for y_2 : 0.04251537935748462

Substring $y_3 =$
MPXFIHNKPJLPJTAWLRSMIJDPVZRDAVBMPVWWLUWWEUOEEIEKMKEDJPFZNFQGCUBASXJWWUJMLVYIZKU
VEEVAKHRGDPBPBKPEIKNFGCKVYUHGTEQOODWRTQJJUXORIDAKRDRVWVBTPYGNLKYDEMAZLSTIWAUZVV
CESLUHLZDKVVJAKLOLWSYVIXWRGEVTPOHMPZMELWZZIQFEEC
Index of Coincidence for y_3 : 0.04155844155844156

Substring $y_4 =$
UUFWHMLFMKFLEFWEOWPMMASHHYZAELYVNCKKVDXQVWADFLPCVILUXWFMNKRHFYJWHNZGGJGYSJKUW
TXLJJLPEDWTRBQALLSZBALQOCRBZARGDUMEJHXZTAKUSIELICWRYEEKZLDHLJJOWVAETJWDLPPKMEZWP
LVWDWXGNRCKLWVSIIDAPCKMBXKINRPEVQWYUKAZWXMEZXJQIZ
Index of Coincidence for y_4 : 0.04160401002506266

Substring $y_5 =$
PVCDFOCXUBCTQVEYMNEKVPNAESXHNZVDWVSZTBWJOSYDAAGHRPZCEGHSGMMAMHPZNBZVXVWTQJAZTLDB
UDWRLVBILRRYYAOWZIDZZNWVIYIMBEDTPIBNWKQLKDWJWEDJVPDFMZTJDDRZTLIZKLLFIZHFHLJEAAGH
XXIKOKDWNGBKSREPCMLHFRSVWVEERRTKYWJAUIKLHIOZWHVFK
Index of Coincidence for y_5 : 0.03886990202779676

Substring $y_6 =$
TYJIWVQEVBAVLLZIMKGZSDVVIUVIRRIOKQZPYBASUXPLBRUWGNEWUJMPJBNJBVLZJAVHIZTAYWWMFZ
LPZHVN YCEDWHPVIOZGMZVWPGVIZMBILUVGYIGGPXSZVKZEIOVMKZWXPQPFHPWEQELHXLVWCLDILZBOY
LYVFAPVSCDYKJSWWWDTAGVEYTWAMQVECVDSRYPSEKUUWEKIKZB
Index of Coincidence for y_6 : 0.04529505582137161

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For  $m = 7$ :  
Substring  $y_1 =$   
LJXWFDTLJFTVLKWFEGWSDSSUWVVLWKLJLVJSUKEDGUGJELGUQWFKMJSRSWVVP SMGAWUUUTDZULHELDZG  
VAWZSEN LWGLWUAELGKOJWGMGJUJZWLAWDZUWWLJFSWLWKLWNZDFLKTUZGYGLLWKVJJVLSSSWLAHGLWXX  
VMSWOVYSJDLZLULUFJVZ  
Index of Coincidence for  $y_1$ : 0.071260086902545

Substring  $y_2 =$   
DPFDWWAPKCAOUAONKSDJAEIPNAEDOKNWKWASAOAARKBKIEIDLNNANKQAJZACWJJJYJAKWUPAAKPIEAAB  
QOOAPKANPNKYZBIEROENGOJJKWKAYDCPKPWWZDPKJNOZEDTAWHCDJAEA O EJS DOPA EKKKJYZODLAJDXBO  
APHPEKD JIEAKKPUQXHKJ  
Index of Coincidence for  $y_2$ : 0.071260086902545

Substring  $y_3 =$   
MUCIVTKMBBZMWMVMIMAVXADAVICTWKBBAMUWABMVMVCWBVZMMBBZBLBJZMUGQAMKZUBLESALBCWTPB  
AIJVBZWEIVBMCWQMBIQMADVDLIIVMOPVKJQGBGJIQWQMJJZWEMWMAEWAUKAZUDCBKIPLIMTBKIMWW  
JETPTVWLMKIEXTIZWICW  
Index of Coincidence for  $y_3$ : 0.06815642458100558

Substring  $y_4 =$   
UVJJUNFUVYKTWYITKMMNVJIRZZRKJVCTYRLEVDLFIKCEEYFFMJFFUYNZPRWGTTJJIKDZVZVJVYEEPRY  
VGFKZZVEFKRIMYEEUIYEETTKZEZRLJVTPRETPINYJLEJEETIVEYIKEZYVCVIFZPZRGYIGKICDVZVKVTD  
RVRRCEZU USSIZEZKNIC

Index of Coincidence for y4: 0.061390440720049656

Substring y5 =

PYEMHLXENPJFEZGSPDPPEZYNRHRPNZPPLWQYPYCEPPNTDPWGPALCFZLDLWZWZLPYJLFZQEJLNPTDTYP  
CAESDZWTCYZTPTDPLERZLSOZOED00SDEZPLOPLTWONDEFDRTPJLZPVLXVYYLNNPLASPFNYCSYFF  
ENREYZPYAPZPZMOESDEZ

Index of Coincidence for y5: 0.06834264432029795

Substring y6 =

TXMIMCQSXPEVLQIRWVSGRVHHIREMVSZWMWXSXXEILRGGGJXSMRSGHPVVWJNVIQXVSTMREL RVVGGWRHI  
XPLIMRIGOIIRGGGTVIAWXQXIIXIEIIXRYMXWRTHZPPXYALXRXSPLIHRVPXYAKHGWCEVJVEWSPSKREXWG  
IEVAIMWJYFYWXIWCSEIK

Index of Coincidence for y6: 0.06225946617008069

Substring y7 =

DRFHVLSLLQLRZLWKZQDPHXVVHBYVQHVUDOLLEFHWWRDUHJGLQRHBLUXHXHPPLYZWQLVXWHBBDRWVQ  
KLZGQRFQVVLHRDULGDHKLPLSUSWELVKRFWEVBDRHYHKVLHLFLLDHWHZWHLVLUXDSJQVWRWVIWPFVYKRK  
EQHHHQWREDWWHUHLXQFB

Index of Coincidence for y7: 0.06790813159528243

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For m = 8:

Substring y1 =

LPCJHCLJCZTEQLESINPVXWAVKPSHLWAEYXEGKMZTJHQNMFCLPSZZWWQLAJZDZEWUKBEDRVVOJKDRFVNV
ITGDGOBERQLJWRDIKDPVTLTRSNIJDLLZHEIJHZGEEIYHDKBIAEVLLBVSVRPTRYMWJEISZIHFHCC

Index of Coincidence for y1: 0.04160283802305894

Substring y2 =

DUJMMVLKBKFLZFKMMDRHUNARRVQJKBLQXLDRCNGUULMJLHBRJJRZIPGYKKFRXZALYTWGQIFSMRLPIR
ZGRLRMEXPGKVZEIVWKPEPRDFJJENAHNWWYPIHZA0AVVKXVECYLVWWDZCEKXAEREIQSIKSPXUUXIIZ

Index of Coincidence for y2: 0.040635330162057566

Substring y3 =

MVEIOTPBYYJVRWNVMPGSPDZHMVLWBRWSLEAMKPGDGDZMAGHSABPWVPMJMILRVDA0AVPGRZBAGEIQNNEKY
RHEEQILWKMJDEOELAPORZWBKGLLOYLWAZETLUWAMCXAUKADGPRJRSOMVYRYWOJVR0AHYJMUZWHLQWNEK

Index of Coincidence for y3: 0.04377973071031202

Substring y4 =

UYMHDAMVPELKOMKPSDSIAZREYKNKYLXUKABIPGRLIVFSPRJQLZLNHSJAJJMLTPSJNGDDABVALGEAWFCI
LAIWUGAHGJAZZIELCMPEXVLBJWXVLTJVRPDTEBVL0LWLPURDVSZIDNKVWBWRNVHVDLUPYUWUZKDIB

Index of Coincidence for y4: 0.04248972022897686

Substring y5 =

PXFFWKUNQLAMTSVQDEXRNRBWKWTPWOU0AFERWEEBFRQMKBNDJXPJQJYTQUUEVLVRLAPYCPZSKZECIO
UBCEPIDWOQK0XWWDVTMEWDGYTPKKDMIDSHKAAVXVLOZZNEVS0YLCFSFLXWDEEH0KWZAFWLPIZSEF

Index of Coincidence for y5: 0.03927813163481953

Substring y6 =

TRWWT0FEXLVUEIPWZSAVIYIHL0TCPMDSAWLCLWJJIWY0MLKNZYLWHVWGTPSWUWLZJVB0ETRPXLZPBVT0VW
ZBEDVGJG0MTSISZYIVSYW0ZQNLPHWEQTLXLLJM0PPLLDAPNCQLJPIWPPGDMTFIQWEVEY0DZKTEESQZ

Index of Coincidence for y6: 0.05038379879144211

Substring y7 =

DXDVNXSLTOWYGRKWJAJZHVVDCVZWUJSUDYI0HGKCEPSGFABUZVXVCUTLVZUKBVERBHIW0PYIKWAI0ZWGVLY
MYTTLONQTLIUJAIJ0RFUWJIPZYLZWEFRLFDWEZVH0GSKZLWJVKIKPSWHJIVFGESPTXPJELKX0ZOCXVJ

Index of Coincidence for y7: 0.041401273885350316

Substring y8 =

JFIULQSFAMWZIWGDMVEHVVEIJNZV0VAMVPEFUBVEDXJWN0FFVUWAMGZXYWAUZQLHLP0CETHQA0VZZI0QGKB
MPGUK0EIEZXSUKLL0XRZWKPOHHW0QEEXFDCMKLRWYJW0FYGSJ0KGSWIAAKKYWKMHPCZWRDAEIMW0LJKW

Index of Coincidence for y8: 0.04131961456802221

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Our outputs verify that m = 7 is the correct guess for the length of the

keyword. For  $m = 6$  and  $m = 8$ , the indices of coincidence for each substring are all somewhat close to 0.038, which is the expected index of coincidence for a completely random substring. By contrast, for  $m = 7$ , all the substrings have indices of coincidence that are close to or exceeding 0.065, which would be the index of coincidence for a substring with letter frequencies approaching those of the English language as a whole. Thus, this indicates that each of the substrings for  $m = 7$  is likely to be a ciphertext encrypted with the same key, which would be expected for a ciphertext encrypted with the Vigenère cipher with a keyword of length 7.